SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**
Question Count: 10

Code of Conduct
I M.Kyathi Sai Priya (192572101) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: __M.Kyathi_

Assessment Tasks
Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1.Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Answer :

Unified Big Data Architecture for a Multinational Retail Organization

A multinational retail organization collects huge amounts of customer data from online shopping websites, mobile applications, physical stores, loyalty programs, social media and payment systems. When this data is stored in separate systems, it creates fragmented customer information. A unified big data architecture can integrate these sources and provide real-time customer analytics and personalized recommendations.

Proposed Architecture

Data Sources → Data Ingestion → Data Lake → Distributed Processing → Analytics/ML → Recommendation Engine → Dashboard

1. Data Sources

Customer data can be collected from:

- Online websites and mobile applications
- Point-of-sale systems in physical stores

- Customer Relationship Management (CRM)
- Loyalty cards
- Social media
- Payment and transaction systems

2. Data Ingestion

Tools such as Apache Kafka can collect both real-time and batch data from different sources.

3. Data Storage

A distributed data lake using HDFS or cloud storage can store structured, semi-structured and unstructured data. Historical customer information can also be maintained in a data warehouse.

4. Data Processing

Apache Spark can process large amounts of customer data in parallel. Real-time processing can identify customer actions immediately.

5. Customer Analytics

Analytics can determine:

- Customer preferences
- Purchase patterns
- Frequently purchased products
- Customer segments
- Customer lifetime value

6. Recommendation System

Machine learning models can recommend products based on:

- Previous purchases
- Browsing history
- Similar customers
- Location

- Current shopping behavior

7. Visualization

Power BI or Tableau dashboards can provide real-time information about sales, customers and product performance.

Benefits

- Creates a 360-degree customer profile
- Provides real-time customer insights
- Improves personalized marketing
- Increases customer satisfaction
- Improves sales and customer retention
- Supports better business decisions

2.Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems

Answer :

Distributed Computing for Large-Scale Scientific Research Data :

Scientific research generates extremely large datasets from areas such as genomics, astronomy, climate science, medical research and particle physics. A centralized system may take a very long time to process such data. Distributed computing and parallel processing provide an efficient solution.

Distributed Computing :

Distributed computing divides a large problem into smaller tasks and processes them simultaneously using multiple computers connected through a network.

Architecture
Large Dataset → Data Partitioning → Multiple Processing Nodes → Parallel Processing → Combined Result

Technologies such as Hadoop and Apache Spark support distributed processing.

Parallel Processing
In a centralized system, one computer processes the entire dataset sequentially.

In a distributed system:
Dataset = D1 + D2 + D3 + D4
Different processing nodes process D1, D2, D3 and D4 simultaneously.

Advantages
1. Reduced Processing Time

Multiple processors work simultaneously, reducing the time required to complete complex calculations.

2. Scalability

More computers can be added when the dataset grows.

3. Fault Tolerance

Distributed systems can replicate data, allowing processing to continue even when one node fails.

4. Better Resource Utilization

CPU, memory and storage resources of multiple systems are utilized.

5. Large Dataset Processing

Petabytes of scientific data can be processed efficiently.

6. Cost Effectiveness

Organizations can use clusters of commodity hardware instead of a single expensive supercomputer for many workloads.

Example

In genomic research, millions of DNA sequences can be divided among different processing nodes. Each node analyzes a portion of the sequences simultaneously, and the results are combined.

Centralized vs Distributed :

Centralized System	**Distributed System**
Single processing system	Multiple processing systems
Limited scalability	Highly scalable
Longer processing time	Faster parallel processing
Single point of failure	Fault tolerant
Limited data capacity	Handles huge datasets

3.Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Answer :

Fraud Detection Framework for a Digital Payment Platform

Digital payment systems generate millions of transactions every day. Fraudsters may perform unusual transactions using stolen accounts, devices or payment information. A big data fraud detection framework can analyze transactions in real time and identify suspicious activities.

Framework

Payment Transactions → Kafka → Real-Time Processing → Feature Extraction → ML Model → Fraud Score → Alert/Block

1. Data Sources

The system collects:

- Transaction amount
- Transaction time
- Customer ID
- Location
- Device information
- Merchant information
- IP address
- Previous transaction history

2. Data Streaming

Apache Kafka can continuously receive transaction events. This allows the system to analyze transactions as soon as they occur.

3. Real-Time Processing

Apache Spark Streaming processes incoming transaction data and extracts important features.

For example:

- Number of transactions in the last 10 minutes
- Unusual transaction amount
- Sudden change in location
- New device usage
- Multiple failed transactions

4. Predictive Model

Machine learning algorithms such as:

- Random Forest
- Decision Tree

- XGBoost
- Neural Networks

can classify transactions as legitimate or fraudulent.

5. Fraud Score

Each transaction can receive a fraud probability.

For example:

Transaction → Fraud Probability = 95% → High Risk → Block/Verify

6. Alert System

High-risk transactions can generate immediate alerts to customers and fraud analysts.

Advantages

- Real-time fraud detection
- Faster response
- Reduced financial losses
- Detection of complex fraud patterns
- Improved prediction accuracy
- Continuous model improvement

4.Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Answer :

Challenges of Multimedia Data in Big Data Environment

Multimedia data includes images, audio, video and other media files. Modern organizations generate huge quantities of multimedia data through cameras, social media, surveillance systems and streaming platforms.

Challenges :

1. Huge Storage Requirements
High-resolution videos and images require significantly more storage than traditional text data.

2. Different Formats
Multimedia can exist in formats such as MP4, AVI, JPEG, PNG, MP3 and WAV.

3. High Processing Requirements
Processing video and audio requires significant CPU/GPU resources.

4. High Bandwidth
Transferring large multimedia files requires high network bandwidth.

5. Complex Searching
Finding specific content inside videos or images is difficult without metadata and indexing.

6. Security and Privacy
Videos and images may contain sensitive personal information.

Suitable Technologies :-

HDFS:
Provides distributed storage across multiple nodes.

Cloud Object Storage:
Amazon S3, Azure Blob Storage and similar systems are suitable for large multimedia files.

Apache Spark:
Can process large-scale multimedia metadata and analytical workloads.

Apache Kafka:
Can handle real-time multimedia event streams.

NoSQL Databases:
Can store metadata associated with multimedia files.

Machine Learning:
Computer vision and speech-processing models can analyze images, audio and video.

Recommended Architecture

Camera/Media Source → Kafka → Data Lake/Object Storage → Spark → ML Processing → Metadata Database → Analytics

5.Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

Answer :

Disaster Recovery and Business Continuity for Cloud Big Data

A cloud-based big data platform stores critical business information. Hardware failures, cyberattacks, network failures or natural disasters can interrupt services. Therefore, a disaster recovery and business continuity strategy is essential.

Disaster Recovery Architecture
Primary Cloud Region → Data Replication → Secondary Region → Automated Failover → Continuous Service

1. Redundancy

Multiple servers and storage devices should be used instead of depending on a single component.

2. Data Replication

Important data should be replicated across different servers or geographical locations.

For example:

Primary Database → Replica Database

If the primary database fails, the replica can be activated.

3. Automated Backup

Regular backups should be performed and stored separately from the primary system.

4. Automated Failover

If the primary server becomes unavailable, the system automatically redirects users to a secondary server.

5. Load Balancing

Traffic is distributed across multiple servers, preventing a single server from becoming overloaded.

6. Monitoring

Cloud monitoring tools continuously monitor:

- CPU
- Memory
- Storage
- Network
- Server availability

7. Recovery Testing

Organizations should regularly test backup restoration and failover procedures.

Benefits:

- Minimizes downtime
- Prevents data loss
- Provides high availability
- Improves fault tolerance
- Supports continuous business operations
- Protects against disasters

6.Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Answer:

Impact of Data Quality Issues in Big Data Analytics

Data quality is essential for reliable big data analytics. Large-scale systems collect information from multiple sources, increasing the possibility of inconsistent, duplicated and missing data.

Major Data Quality Problems:

1. Inconsistency

Different systems may represent the same information differently.

Example:

- India
- IND
- IN

All may refer to the same country.

2. Duplication

The same customer may be stored multiple times, causing incorrect customer counts and sales analysis.

3. Missing Values

Important attributes such as age, address or transaction amount may be missing.

4. Incorrect Data

Incorrect values can result from manual entry errors or faulty sensors.

Impact

Poor data quality can cause:

- Incorrect reports
- Wrong business decisions
- Inaccurate machine learning predictions
- Incorrect customer segmentation
- Financial losses
- Reduced trust in analytics

Data Governance Strategies :

1. Data Validation
Validate data before storing it.

2. Data Cleaning
Correct errors and standardize formats.

3. Deduplication
Identify and remove duplicate records.

4. Missing Value Handling
Use suitable techniques such as mean, median or model-based imputation.

5. Master Data Management
Maintain consistent definitions for important entities such as customers and products.

6. Metadata Management
Maintain information about data sources, definitions and ownership.

7. Data Quality Monitoring
Continuously measure completeness, accuracy, consistency and validity.

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Answer :

Online streaming platforms such as video and music services handle millions of users and large volumes of behavioral data. A scalable recommendation system can analyze this data and suggest content according to individual user preferences.

Architecture :

User Activity → Kafka → Data Lake → Spark → Recommendation Model → Recommendation API → User

1. Data Collection

The system collects:

- Watch history
- Search history
- Likes and dislikes
- Ratings
- Skipped content
- Watch duration
- Genre preferences

2. Data Storage

A distributed data lake stores large-scale user behavior and content information.

3. Distributed Processing

Apache Spark processes data from millions of users simultaneously.

4. Recommendation Algorithms

Collaborative Filtering:

Recommends content based on users with similar preferences.

Content-Based Filtering:

Recommends content similar to what the user previously watched.

Hybrid Recommendation:

Combines collaborative and content-based techniques.

5. Real-Time Recommendations

When a user watches a new movie, the system can update the user's profile and generate new recommendations.

Benefits

- Personalized content
- Increased user engagement
- Increased watch time
- Better customer satisfaction
- Reduced content-search time
- Scalable to millions of users

8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Answer :

Role of Cloud Computing in Modern Big Data Ecosystems

Cloud computing has become an important part of modern big data because organizations need large amounts of computing power and storage.

Role of Cloud Computing

Cloud platforms provide:

- Computing resources
- Distributed storage
- Databases
- Data processing
- Machine learning services
- Analytics platforms

Advantages :

1. Scalability

Resources can be increased when data volume increases and reduced when demand decreases.

2. Cost Management

Organizations can use a pay-as-you-go model instead of purchasing expensive infrastructure.

3. Flexibility

Organizations can choose different computing, storage and analytics services according to requirements.

4. High Availability

Cloud providers offer geographically distributed infrastructure and backup mechanisms.

5. Faster Deployment

Big data applications can be deployed quickly without installing physical servers.

Limitations :

1. Security

Sensitive information stored in cloud environments can face security threats.

2. Privacy

Organizations must ensure that personal and confidential data is properly protected.

3. Vendor Lock-In

Moving applications from one cloud provider to another may be difficult.

4. Cost Overruns

Poor resource management can lead to unexpected cloud bills.

5. Internet Dependency

Cloud services generally require reliable network connectivity.

9.Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Answer :

Real-Time Analytics Framework for Transportation and Logistics

Transportation and logistics organizations generate continuous data from GPS devices, vehicles, traffic systems, fuel sensors, weather services and delivery applications. Big data analytics can process this information in real time to improve transportation operations.

Framework

GPS/IoT Sensors → Kafka → Spark Streaming → Analytics/ML → Decision Engine → Dashboard

1. Data Collection

Sensors collect:

- Vehicle location
- Speed
- Fuel consumption
- Engine condition
- Traffic information
- Delivery status

2. Data Streaming

Apache Kafka collects continuous real-time sensor and GPS data.

3. Real-Time Processing

Spark Streaming analyzes incoming data and detects traffic congestion, delays and abnormal vehicle conditions.

4. Route Optimization

Analytics can identify the most efficient route based on:

- Traffic
- Distance
- Weather
- Road conditions
- Delivery priority

5. Fuel Efficiency

Fuel consumption data can be analyzed to identify:

- Unnecessary idling
- Inefficient routes
- Excessive speed
- Poor driving patterns

6. Predictive Maintenance

Machine learning can predict possible vehicle failures before they occur.

7. Dashboard

Managers can monitor:

- Vehicle locations
- Delivery status
- Fuel consumption
- Traffic conditions
- Estimated arrival times

Benefits

- Reduced travel time
- Lower fuel consumption
- Improved route planning
- Reduced maintenance costs
- Faster decision-making
- Improved delivery performance

10.Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust

Answer :

Ethical and Legal Challenges of Big Data

Big data enables organizations to collect and analyze massive amounts of information. \

However, improper use of personal data can create privacy, security, ethical and legal problems.

Major Challenges

1. Privacy

Organizations may collect sensitive personal information such as location, purchasing behavior and online activity.

2. Lack of Consent

Users may not always understand what information is collected or how it is used.

3. Data Security

Large databases are attractive targets for hackers and cybercriminals.

4. Algorithmic Bias

Biased training data can cause machine learning systems to make unfair decisions.

5. Transparency

Users may not understand how automated systems make decisions about them.

6. Data Ownership

Organizations need clear policies regarding who owns and controls collected data.

7. Legal Compliance

Organizations must follow applicable data-protection and privacy laws and regulations.

Organizational Policies

1. Obtain Consent
Users should be clearly informed before collecting personal information.

2. Data Minimization
Only necessary information should be collected.

3. Encryption
Sensitive data should be encrypted during storage and transmission.

4. Access Control
Only authorized employees should access sensitive information.

5. Anonymization
Personal identifiers should be removed where possible.

6. Bias Monitoring
AI models should be regularly tested for unfair or discriminatory outcomes.

7. Data Retention Policy
Data should not be stored longer than necessary.

8. Regular Auditing
Security and compliance audits should be conducted regularly.